

Motor Datasheet

Operating Point: Continuous | Temperature: 20 °C

Geometry (active elements only)

Parameter	Value	Unit
Machine Outer Diameter	113.0	mm
Machine Inner Diameter	51.7	mm
Machine Thickness / Width	26.0	mm

Mass Properties (active elements only)

Parameter	Value	Unit
Machine Mass	1.27	kg
Rotor Mass	444.2	g
Stator Mass	826.2	g
Rotor Moment of Inertia (active elements only)	8437.0	g cm2

Electrical Properties

Parameter	Value	Unit
Supply Current ('RMS')	45.0	A
Phase Resistance	10.5	mOhm
D-Axis Inductance (Ld)	29.9	μH
Q-Axis Inductance (Lq)	32.3	μH
PM Flux Linkage	9.2	mWb
Frequency at operating point (electrical)	416.7	Hz
Winding Configuration	wye (star)	-
DC Supply Voltage	48.0	V

Performance

Parameter	Value	Unit
Torque (Continuous)	7.62	Nm
Speed	2500.0	rpm
BackEMF Constant (Ke)	9.6	mV/rpm
Motor Constant (Km, only resistive losses)	955.4	mNm/sqrt(W)
Specific Motor Constant (Km / mass)	752.1	mNm/sqrt(W)/kg
Torque Constant (Kt, no-load)	140.8	mNm/A_peak
Torque Ripple	3.1	%
Saturation	14.9	%

Losses

Parameter	Value	Unit
Total Losses	76.5	W
Resistive Losses	63.6	W

Configuration

Parameter	Value	Unit
Number of Stators	1	-
Number of Rotors	1	-
Number of Slots	24	-
Pole Pairs	10	-
Number of Parallel Paths	1	-
Number of Turns per Tooth (designed incl. parallel paths)	4	-
Number of Phase Turns (all phases in series)	32	-

Materials

Component	Material	
Stator Core	N030-Cogent	-
Magnets	BMN-52SH (GBD)	-

3D Model

